

Stopped Experiments in Stochastic Bandit Convex Optimization

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Abstract

We study stochastic bandit convex optimization through stopped experiments. At a posterior state and risk scale b , the pointwise stopped cost is the least scale-adjusted regret of an adaptive experiment, divided by the probability that the experiment reduces posterior Bayes risk by a fixed factor. Unsuccessful branches may be abandoned; successful runs stop at the first contraction. The value therefore retains the observation-dependent continuation of the experiment without assigning each observation an information budget.

For repeated decisions with a fixed latent environment, a uniform stopped complexity bound SC gives $O(1 + \sqrt{SCT})$ Bayes regret, while a pointwise value c gives an $\Omega(\sqrt{cn})$ lower bound at its calibrated horizon. Neither direction requires convexity or Gaussian feedback. For bounded convex losses with Gaussian observations, the pointwise experiments can be chosen measurably, and worst-prior Bayes regret equals minimax regret. The polynomial dimension exponents consequently satisfy $\alpha_{\text{reg}} = \alpha_{\text{stop}}/2$, without an additional matching assumption.

For dimple losses at scale d^{-1} , nonadaptive batches and stopped fresh-direction experiments require $\Omega(d^5)$ queries, whereas an observation-adaptive experiment uses $\tilde{O}(d^4)$. At horizon d^5 , the corresponding regret bounds are $\Omega(d^4)$ and $\tilde{O}(d^{7/2})$. This separation shows that observations can be valuable chiefly because they orient later queries.

1 Introduction

Let $C \subset \mathbb{R}^d$ be a compact convex body. An unknown continuous convex loss $F : C \rightarrow [0, 1]$ is fixed throughout the interaction. At round t , the learner chooses $X_t \in C$, observes

$$O_t = F(X_t) + \xi_t, \quad \xi_t \stackrel{\text{iid}}{\sim} N(0, 1),$$

and incurs regret $F(X_t) - \min_C F$. The cumulative regret is

$$R_T = \sum_{t=1}^T \{F(X_t) - \min_C F\}.$$

In the nontrivial long-horizon regime, the best general upper bound is $\tilde{O}(d^{3/2}\sqrt{T})$ (Lattimore and György, 2023; Fokkema et al., 2024). A complementary simple-regret formulation and a particularly direct localization method for noisy zeroth-order convex optimization are developed by Carpentier (2025). The recent lower bound $\tilde{\Omega}(d^{5/4}\sqrt{T})$ of Rajaraman (2026) is the first for the full class with dimension dependence strictly larger than the linear-bandit rate. Neither endpoint is known to be sharp.

The local quantity. One obstacle is intrinsically multistep: an observation may be valuable mainly because it determines where the next probe should be placed. A one-query quantity may therefore price the first direction before its observation-dependent continuation is known.

We measure the latter directly. Let Q_t be the posterior after t queries and

$$r(Q_t) = \inf_{x \in C} \mathbb{E}[F(x) - \min_C F \mid H_t]$$

its Bayes risk. At a state where $r(Q)$ is of order b , run an adaptive experiment until it either aborts or first reaches $r(Q_t) \leq \gamma b$. If N is its duration and $\mathbf{M}$ the successful first-hit event, then $\mathbb{E}R_N/\mathbb{P}(\mathbf{M})$ is expected regret per unit probability of contraction. Its scale-normalized form,

$$\frac{b \mathbb{E}R_N}{\mathbb{P}(\mathbf{M})}.$$

is the *pointwise stopped cost* at (Q, b) . The factor b gives it the units of squared regret. No dimension factor, auxiliary random target, or information budget is inserted.

This definition also needs no separate query penalty. Every query executed before the hit has conditional expected regret at least γb , and hence

$$\mathbb{E}R_N \geq \gamma b \mathbb{E}N.$$

Consequently, adding a query charge $b^2 \mathbb{E}N$ changes the value by at most the fixed factor $1 + \gamma^{-1}$.

A two-sided reduction. Let $\text{SC}^{\text{pt}}(d, B)$ be the largest pointwise stopped value over posterior states and risk scales $B^{-1} \leq b \leq 1$, and let $\text{SC}^{\text{unif}}(d, B)$ be the smallest uniform bound achieved by one measurable state-dependent experiment rule. We prove

$$\mathfrak{R}^{\text{B}}(d, T) \leq \inf_{T^{-1} \leq \eta < 1/2} \left\{ C_{\gamma, \vartheta} \frac{\text{SC}^{\text{unif}}(d, T)}{\eta} + 2T\eta \right\}, \quad (1)$$

$$\mathfrak{R}^{\text{B}}(d, n) \geq \frac{\gamma c b n}{c + \gamma b^2 n} \quad (2)$$

for every $c < \text{SC}^{\text{pt}}(d, B)$, at some initial posterior and scale $b \in [B^{-1}, 1]$. The upper bound concatenates stopped experiments over geometric risk scales. The lower bound is more elementary: before a hit the policy pays at least γb per round, while producing a hit with probability p costs at least cp/b regret. Balancing the two possibilities gives (2). There is no terminal recommendation, convex averaging, or online-to-batch step.

At $n \asymp c/b^2$, the lower bound is $\Omega_{\gamma}(\sqrt{cn})$. The two finite-value statements do not assert a same-horizon equality: the lower horizon is calibrated to the scale of a hard state. They do give an exact dimension-exponent identity once the pointwise and uniform values are measurably identified.

Pointwise and uniform values. The two-sided reduction itself is decision-theoretic: it uses neither convexity nor Gaussian noise. Those assumptions enter when we specialize the framework to BCO and verify all implementation and minimax requirements without placing them among the premises. For compact C , Borel loss classes, and Gaussian feedback, a finite-cap stopped experiment is a belief-state control problem with Borel primitives. Universally measurable near-minimizers and monotone truncation identify the two values:

$$\text{SC}^{\text{pt}}(d, B) = \text{SC}^{\text{unif}}(d, B) =: \text{SC}(d, B).$$

A direct strategic-measure argument also gives finite-horizon Bayes–minimax equality. If α_{stop} is the polynomial dimension exponent of $\text{SC}(d, d^m)$, uniformly over polynomial scale windows, and α_{reg} is the corresponding exponent of minimax regret normalized by $\sqrt{T}$, then

$$\alpha_{\text{reg}} = \frac{\alpha_{\text{stop}}}{2}.$$

The identity holds for every bounded Gaussian BCO subclass, with no nondegeneracy, entropy-matching, or localization premise. It is an exact exponent reduction, not a same-horizon equality and not, by itself, a new regret bound.

Observation-adaptive continuation. The exponent reduction becomes informative when the stopped problem can be bounded more sharply than one-query restrictions. For the dimple losses of [Bakhtiari et al. \(2025\)](#), at $\varepsilon = d^{-1}$ we prove

$$\text{SC}_b^{[1]}, \text{SC}_b^{\text{fresh}} = \Omega(d^3), \quad \text{SC}_b = \tilde{O}(d^2).$$

The fresh class may adapt every radius but must draw a new uniform direction at each query. The unrestricted experiment first creates a weak direction, uses observation-adapted tangent differences to amplify it, and then contracts the angular error geometrically. It requires $\tilde{O}(d^4)$ queries, compared with $\Omega(d^5)$ for fresh directions, and yields a family-specific $\tilde{O}(\min\{T/d, d\sqrt{T}\})$ regret bound. This is a separation between experiment classes, not a lower bound against all algorithms for the dimple family.

Information and statistical testing. The stopped cost is a decision cost, not an information measure. Mutual information, Le Cam, Fano, Assouad, and interactive change of measure remain useful ways to bound the cost of reaching the lower-risk set; the fresh-direction lower bound uses mutual information in precisely this way. Maximal information gain may likewise enter model-specific estimates of $\text{SC}_b(Q)$, but it need not appear in the definition. A stopped experiment carries the complete posterior forward, so partial progress matters through the continuation state.

Related work. Information ratios, AIR, and DEC turn one-step regret–information or decision–estimation comparisons into sequential regret guarantees ([Russo and Van Roy, 2016, 2018](#); [Lattimore and György, 2021](#); [Foster et al., 2021, 2022, 2023](#); [Xu and Zeevi, 2025](#); [Neu et al., 2024](#)). Chained ratios organize information across action scales ([Gouverneur et al., 2024](#)). These methods can re-optimize after every observation; our distinction concerns the local comparison being optimized, not whether the resulting algorithm is sequential.

Classical sequential design jointly optimizes experiments and stopping ([Chernoff, 1959](#); [Naghshvar and Javidi, 2013](#)). Interactive Le Cam, Fano, and Assouad inequalities treat adaptive transcript laws and multi-step interactions directly ([Chen et al., 2024](#)). The present stopping event is decision-specific: it is the first constant-factor reduction in posterior Bayes risk, and the experiment is charged cumulative regret before reaching that event.

Near-optimal targets and localization appear in rate–distortion, satisficing, and scale-sensitive ratio analyses ([Dong and Van Roy, 2018](#); [Russo and Van Roy, 2022](#); [Arumugam and Van Roy, 2021](#); [Bubeck and Sellke, 2023](#)). A distinct modern line develops data-dependent localization through pointwise complexities ([Li and Xu, 2026](#); [Xu, 2026c,b](#)). There “pointwise” refers to localization around a realized input, distribution, or geometric point. Our pointwise value instead permits the best stopped experiment to be chosen separately at each posterior state; the Gaussian implementation theorem asks whether all these choices can be assembled into one measurable rule. The two forms

of localization are complementary rather than identical. Polynomial gains from measurements that orient later probes arise in structured best-arm identification (Maiti et al., 2026). The Bellman-sufficient framework of Xu (2026a) is broader: it treats general stateful sequential problems, and its upper and lower bounds are linked through a chosen information index. Its information-risk sandwich matches a log-penalized Bellman upper value with a Bellman-Fano lower value when their information scales align. The present paper studies the repeated-decision specialization in which the posterior is sufficient and the one-step gap is nonnegative. It gives up the information index in exchange for one operational stopped value on both sides. Gaussian implementation makes the resulting exponent comparison unconditional, but the result is narrower than the stateful framework of Xu (2026a). Thus the former is a same-horizon information comparison under alignment conditions, whereas the latter is a cross-scale stopped reduction that becomes exact at the level of polynomial dimension exponents. On posterior space, its recursion is a stochastic shortest-path problem (Eaton and Zadeh, 1962; Bertsekas and Tsitsiklis, 1991; Russo et al., 2018); the new part is the decision-risk stopping event and its two-sided connection to regret.

2 Stopped contraction cost

2.1 Repeated decisions with a fixed latent environment

We first work beyond convex optimization, but within a repeated decision problem with a fixed latent environment. This is the static-state specialization of the Bellman-sufficient framework of Xu (2026a): the pre-action state is the current posterior, the same action problem recurs, and the action gap below is the nonnegative one-step Bellman loss. General stateful control may also admit a Bellman-sufficient state, but it need not have the persistence property used to join successive risk scales.

Let Ω be a standard Borel environment space, $\mathcal{A}$ an action space, and $\Delta_\omega(a) \in [0, 1]$ a nonnegative action gap. Querying a produces an observation from a kernel $P_\omega(\cdot | a)$. The environment $\omega \sim Q$ is drawn once and remains fixed. A policy chooses A_t from the pre-action history H_{t-1} , observes O_t , and incurs $\Delta_\omega(A_t)$. We include all independent exogenous policy randomization in H_0 and write $\mathcal{H}_t = \sigma(H_t)$ for the completed history filtration.

For a posterior Q , define the Bayes risk

$$r(Q) = \inf_{a \in \mathcal{A}} \mathbb{E}_Q \Delta_\omega(a).$$

We assume the posterior is a sufficient controlled state, r is measurable, and measurable ϵ -Bayes actions exist. These are standard in compact Borel models and are verified below for Gaussian BCO. The theorem applies to static latent-model decision problems of the DEC/DMSO type. The nonnegative gap assumption is substantive: with signed one-step increments, stopping can remove compensating future terms and elapsed time need not be paid by regret. Such problems require a separate query charge or a lower-bounded Bellman loss.

The repeated-decision structure implies that posterior Bayes risk is a nonnegative supermartingale. Indeed,

$$\mathbb{E}[r(Q_{t+1}) | \mathcal{H}_t] \leq \inf_{a \in \mathcal{A}} \mathbb{E}[\mathbb{E}\{\Delta_\omega(a) | \mathcal{H}_{t+1}\} | \mathcal{H}_t] = r(Q_t). \quad (3)$$

This fact, rather than convexity or a particular observation law, controls returns to a risk scale in the upper bound.

Fix

$$0 < \gamma < \vartheta < 1. \quad (4)$$

A pair $s = (Q, b)$ is *active* when

$$\vartheta b < r(Q) \leq b.$$

Starting from an active state, let

$$\tau_b = \inf\{t \geq 1 : r(Q_t) \leq \gamma b\} \quad (5)$$

be the first posterior-risk contraction.

Definition 2.1 (Stopped contraction cost). *An admissible experiment consists of an adaptive query policy and a positive, possibly infinite, $(\mathcal{H}_t)$ -stopping time σ . It runs for*

$$N = \sigma \wedge \tau_b$$

rounds and succeeds on $\mathbf{M} = \{\tau_b \leq \sigma\}$. We require $N < \infty$ almost surely. Expectations below are initially extended-valued; Lemma 2.2 shows that every experiment with finite displayed ratio has $\mathbb{E}N < \infty$. Define

$$\text{SC}_b(Q) := \inf_{\pi, \sigma} \frac{b \mathbb{E}_Q R_N}{\mathbb{P}_Q(\mathbf{M})}, \quad R_N = \sum_{t=1}^N \Delta_\omega(A_t), \quad (6)$$

with ratio $+\infty$ when the success probability is zero.

The first-hit convention loses nothing: after reaching the lower-risk set, additional observations cannot improve the success event and only add nonnegative regret. Voluntary stopping is essential, since an experiment may abandon an unpromising branch.

Lemma 2.2 (Query cost is already charged). *Every admissible experiment satisfies*

$$\mathbb{E}_Q R_N \geq \gamma b \mathbb{E}_Q N. \quad (7)$$

Consequently,

$$\text{SC}_b(Q) \leq \inf_{\pi, \sigma} \frac{b^2 \mathbb{E}_Q N + b \mathbb{E}_Q R_N}{\mathbb{P}_Q(\mathbf{M})} \leq (1 + \gamma^{-1}) \text{SC}_b(Q).$$

Proof. The event $\{t \leq N\}$ lies in $\mathcal{H}_{t-1}$, and on this event the pre-query posterior has not hit the lower-risk set. If π_t is the randomized action kernel, then

$$\mathbb{E}[\Delta_\omega(A_t) \mid \mathcal{H}_{t-1}] = \int \mathbb{E}_{Q_{t-1}} \Delta_\omega(a) \pi_t(da \mid H_{t-1}) \geq r(Q_{t-1}) > \gamma b.$$

Tonelli's theorem applied to the stopped sum proves (7); the comparison follows immediately. $\square$

2.2 Pointwise and uniform values

Let $\mathfrak{D} = (\mathfrak{D}_d)$ be a sequence of such decision problems. For a scale window $B^{-1} \leq b \leq 1$, write $\mathbf{S}_\mathfrak{D}(d, B)$ for the active posterior states. Define

$$\text{SC}_\mathfrak{D}^{\text{pt}}(d, B) = \sup_{(Q, b) \in \mathbf{S}_\mathfrak{D}(d, B)} \inf_{\mathcal{E}} \frac{b \mathbb{E}_Q R_N}{\mathbb{P}_Q(\mathbf{M})}, \quad (8)$$

$$\text{SC}_\mathfrak{D}^{\text{unif}}(d, B) = \inf_{\Sigma \text{ universally measurable}} \sup_{(Q, b) \in \mathbf{S}_\mathfrak{D}(d, B)} \frac{b \mathbb{E}_Q^{\Sigma(Q, b)} R_N}{\mathbb{P}_Q^{\Sigma(Q, b)}(\mathbf{M})}. \quad (9)$$

The supremum of an empty set is zero. Always $\text{SC}^{\text{pt}} \leq \text{SC}^{\text{unif}}$. The first quantity chooses the best experiment separately at each state. The second requires all statewise choices to form one universally measurable experiment rule.

Let

$$\mathfrak{R}_{\mathfrak{D}}^{\text{B}}(d, T) = \sup_Q \inf_{\pi} \mathbb{E}_Q^{\pi} R_T$$

denote worst-prior Bayes regret, with any outer supremum over known action spaces understood.

Theorem 2.3 (Two-sided stopped reduction). *For every $T \geq 2$ and $T^{-1} \leq \eta < 1/2$,*

$$\mathfrak{R}_{\mathfrak{D}}^{\text{B}}(d, T) \leq \min \left\{ T, C_{\gamma, \vartheta} \frac{\text{SC}_{\mathfrak{D}}^{\text{unif}}(d, T)}{\eta} + 2T\eta \right\}. \quad (10)$$

In particular,

$$\mathfrak{R}_{\mathfrak{D}}^{\text{B}}(d, T) \leq C_{\gamma, \vartheta} \min \left\{ T, 1 + \sqrt{\text{SC}_{\mathfrak{D}}^{\text{unif}}(d, T) T} \right\}. \quad (11)$$

Conversely, for every $0 < c < \text{SC}_{\mathfrak{D}}^{\text{pt}}(d, B)$, some active initial posterior and scale $B^{-1} \leq b \leq 1$ satisfy, for every $n \geq 1$,

$$\mathfrak{R}_{\mathfrak{D}}^{\text{B}}(d, n) \geq \frac{\gamma c b n}{c + \gamma b^2 n} \geq \frac{1}{2} \min \left\{ \gamma b n, \frac{c}{b} \right\}. \quad (12)$$

If $c/(\gamma b^2) \geq 2$, then at

$$n = \left\lfloor \frac{c}{\gamma b^2} \right\rfloor$$

one has

$$\mathfrak{R}_{\mathfrak{D}}^{\text{B}}(d, n) \geq \frac{\sqrt{\gamma}}{4} \sqrt{c n}. \quad (13)$$

Proof. If $\text{SC}_{\mathfrak{D}}^{\text{unif}}(d, T) = +\infty$, the upper statement is the trivial bound T . Otherwise fix one universally measurable selector Σ whose worst-state ratio is at most $\text{SC}_{\mathfrak{D}}^{\text{unif}}(d, T) + \epsilon$. For the upper bound, use geometric risk buckets $b_j = \vartheta^j$. Whenever $r(Q_t) > 2\eta$, invoke $\Sigma(Q_t, b_j)$ at the unique active scale; otherwise play a measurable δ_t -Bayes action, where $\sum_t \delta_t \leq \delta$. Virtually complete the last block if it crosses the horizon. Every actual block makes at least one query, so only this final block can cross T . At scale b_j , the selector's ratio bound gives

$$b_j \mathbb{E}[R_k \mid H_{S_k}] \leq \{\text{SC}_{\mathfrak{D}}^{\text{unif}}(d, T) + \epsilon\} p_k,$$

where p_k is the conditional hit probability.

After the k -th hit in bucket j , at time U_k , one has $r(Q_{U_k}) \leq \gamma b_j$. By (3) and Doob's maximal inequality, applied first over finite horizons after U_k and then by monotone convergence,

$$\mathbb{P} \left\{ \sup_{t \geq U_k} r(Q_t) > \vartheta b_j \mid \mathcal{H}_{U_k} \right\} \leq \frac{\gamma}{\vartheta}.$$

A later return to the same active bucket requires the event inside this probability. The number of hits in one bucket is therefore geometrically dominated and has expectation at most $(1 - \gamma/\vartheta)^{-1}$. Hence

$$\mathbb{E} R_{\text{explore}} \leq \frac{\text{SC}_{\mathfrak{D}}^{\text{unif}}(d, T) + \epsilon}{1 - \gamma/\vartheta} \sum_{b_j \geq 2\eta} \frac{1}{b_j} \leq \frac{\text{SC}_{\mathfrak{D}}^{\text{unif}}(d, T) + \epsilon}{2(1 - \vartheta)(1 - \gamma/\vartheta)\eta}.$$

The non-experiment rounds contribute at most $2T\eta + \delta$. Letting $\delta, \epsilon \downarrow 0$ proves (10); optimizing η , with the boundary choices and the trivial bound T , proves (11).

For the converse, choose an active state with $\text{SC}_b(Q) > c$. Run an arbitrary n -round policy, stop it at $N = \tau_b \wedge n$, and put $p = \mathbb{P}_Q(\tau_b \leq n)$. The stopped-cost premise and nonnegativity give

$$\mathbb{E}_Q R_n \geq \mathbb{E}_Q R_N \geq \frac{c}{b} p.$$

By the same pre-hit argument as in Lemma 2.2,

$$\mathbb{E}_Q R_n \geq \mathbb{E}_Q R_N \geq \gamma b \mathbb{E}_Q N \geq \gamma b n (1 - p).$$

Minimizing the maximum of these two affine functions over $p \in [0, 1]$ gives (12). At the displayed calibrated horizon, $n \geq c/(2\gamma b^2)$, and direct substitution gives $\mathbb{E}_Q R_n \geq c/(4b) \geq (\sqrt{\gamma}/4)\sqrt{cn}$. $\square$

2.3 Bellman continuation

The stopped value is a stochastic shortest-path problem on posterior space. For a query cap L and price $q > 0$, put $W_0^q = 0$ and

$$\begin{aligned} J_k^q(Q, b) &= \inf_{a \in \mathcal{A}} \left\{ b \mathbb{E}_Q \Delta_\omega(a) \right. \\ &\quad \left. + \int [-q \mathbf{1}\{r(Q^{a,o}) \leq \gamma b\} + \mathbf{1}\{r(Q^{a,o}) > \gamma b\} W_{k-1}^q(Q^{a,o}, b)] P_Q(do | a) \right\}, \quad (14) \\ W_k^q(Q, b) &= \min\{0, J_k^q(Q, b)\}. \end{aligned}$$

The root uses J_L^q , so the experiment makes at least one query; W_k^q permits voluntary stopping on a non-hitting branch. If $\text{SC}_{b,L}$ is the infimum in (6) over experiments with $N \leq L$ almost surely, then

$$\text{SC}_{b,L}(Q) < q \iff J_L^q(Q, b) < 0, \quad \text{SC}_b(Q) = \inf_{L \geq 1} \text{SC}_{b,L}(Q). \quad (15)$$

The first equivalence follows directly by subtracting $q\mathbb{P}(\mathbf{M})$ from the stopped cost; the second follows by monotone truncation. Early observations receive value through $Q^{a,o}$ and its continuation, even when the first query does not itself hit.

Remark 2.4 (Information as a method). *Mutual information may help bound the stopped value without defining it. For a target fixed across visits to one scale, a valid entropy or maximal-information-gain budget can amortize non-hitting blocks by the chain rule. Posterior- or horizon-dependent budgets require a separate randomly stopped, multiscale accounting theorem.*

3 Gaussian implementation for bandit convex optimization

Return to stochastic BCO on a known compact convex body C . Let $\mathfrak{F}_d(C)$ be a Borel subset of $\mathcal{C}(C, [0, 1])$, with the uniform topology. Write

$$\mathfrak{R}_{\mathfrak{F}}^{\mathbf{M}}(d, T) = \sup_C \inf_{\pi} \sup_{f \in \mathfrak{F}_d(C)} \mathbb{E}_f^\pi R_T, \quad (16)$$

$$\mathfrak{R}_{\mathfrak{F}}^{\mathbf{B}}(d, T) = \sup_C \sup_{Q \text{ on } \mathfrak{F}_d(C)} \inf_{\pi} \mathbb{E}_Q^\pi R_T. \quad (17)$$

The values $\text{SC}_{\mathfrak{F}, C}^{\text{pt}}$ and $\text{SC}_{\mathfrak{F}, C}^{\text{unif}}$ are defined by (8)–(9); an outer supremum over C gives $\text{SC}_{\mathfrak{F}}^{\text{pt}}$ and $\text{SC}_{\mathfrak{F}}^{\text{unif}}$.

Theorem 3.1 (Measurable implementation and Bayes–minimax equality). *For every finite d, T, B ,*

$$\mathfrak{R}_{\mathfrak{F}}^{\text{M}}(d, T) = \mathfrak{R}_{\mathfrak{F}}^{\text{B}}(d, T), \quad (18)$$

$$\text{SC}_{\mathfrak{F}, C}^{\text{pt}}(d, B) = \text{SC}_{\mathfrak{F}, C}^{\text{unif}}(d, B). \quad (19)$$

The supremum in (18) may be restricted to finitely supported priors. For every $\epsilon > 0$, the second equality is implemented by a universally measurable state-dependent experiment whose finite deterministic cap may depend on the state.

The first equality is a finite-horizon minimax theorem, not an asymptotic interchange. The second resolves the only difference between the numerical pointwise value and an admissible uniform rule. Its proof is simpler for SC than for an information-based value: the capped Bellman recursion (14) contains only a bounded regret cost and a terminal hit reward.

Write their common stopped complexity as

$$\text{SC}_{\mathfrak{F}}(d, B).$$

Define the stopped-complexity and regret exponents

$$\alpha_{\mathfrak{F}}^{\text{stop}} = \sup_{m \in \mathbb{N}} \limsup_{d \rightarrow \infty} \frac{\log\{1 + \text{SC}_{\mathfrak{F}}(d, d^m)\}}{\log d}, \quad (20)$$

$$\alpha_{\mathfrak{F}}^{\diamond} = \sup_{M \in \mathbb{N}} \limsup_{d \rightarrow \infty} \sup_{2 \leq T \leq d^M} \frac{\log\{1 + \mathfrak{R}_{\mathfrak{F}}^{\diamond}(d, T)/\sqrt{T}\}}{\log d}, \quad \diamond \in \{\text{B}, \text{M}\}. \quad (21)$$

Theorem 3.2 (Exact exponent relation). *For every sequence of Borel subclasses of bounded convex losses under Gaussian feedback,*

$$\alpha_{\mathfrak{F}}^{\text{M}} = \alpha_{\mathfrak{F}}^{\text{B}} = \frac{\alpha_{\mathfrak{F}}^{\text{stop}}}{2}. \quad (22)$$

No information-scale matching or nondegeneracy assumption is required.

Proof. For $T \leq d^M$, monotonicity of the scale window and Theorem 2.3 give

$$\frac{\mathfrak{R}_{\mathfrak{F}}^{\text{B}}(d, T)}{\sqrt{T}} \leq C_{\gamma, \vartheta} \left\{ T^{-1/2} + \sqrt{\text{SC}_{\mathfrak{F}}(d, d^M)} \right\}.$$

Taking the exponents proves $\alpha_{\mathfrak{F}}^{\text{B}} \leq \alpha_{\mathfrak{F}}^{\text{stop}}/2$. The general BCO regret bound and the online-to-stop conversion in Remark 3.3 also give $\alpha_{\mathfrak{F}}^{\text{stop}} \leq 3$. Since both exponents are nonnegative, the upper inequality settles the case $\alpha_{\mathfrak{F}}^{\text{stop}} = 0$.

Suppose now that $\alpha_{\mathfrak{F}}^{\text{stop}} > 0$ and fix $0 < \epsilon < \alpha_{\mathfrak{F}}^{\text{stop}}/4$. Choose a fixed m for which the inner limit superior in (20) exceeds $\alpha_{\mathfrak{F}}^{\text{stop}} - \epsilon$. Along a subsequence of dimensions, the definition of the supremum over active states then supplies states (Q_d, b_d) with $b_d \geq d^{-m}$, and numbers

$$d^{\alpha_{\mathfrak{F}}^{\text{stop}} - 2\epsilon} \leq c_d < \text{SC}_{b_d}(Q_d).$$

We first record that their calibrated horizons are polynomial. On every window $b \geq d^{-m}$, and uniformly for $n \leq d^{4+2m}$, the general BCO bounds of Lattimore and György (2023, Theorem 1) and, for arbitrary compact bodies, Fokkema et al. (2024, Theorem 2) give a policy with

$$\mathbb{E}R_n \leq D_d \sqrt{n} + A_d, \quad D_d = d^{3/2+o_m(1)}, \quad A_d = d^{3+o_m(1)}$$

The high-probability arbitrary-body bound yields the same expectation bound by taking a polynomially small failure probability and using $R_n \leq n$. Its logarithmic factors are $d^{o_m(1)}$ uniformly on this polynomial horizon range. Take the least integer n satisfying

$$n \geq \max \left\{ \frac{16D_d^2}{\gamma^2 b^2}, \frac{4A_d}{\gamma b} \right\}.$$

It obeys $n \leq d^{3+2m+o_m(1)}$, and hence lies in the same polynomial range. Then $\mathbb{E}R_n \leq \gamma bn/2$, so the online-to-stop inequality in Remark 3.3 gives, uniformly on the window,

$$\text{SC}_b(Q) \leq 2b(D_d\sqrt{n} + A_d) = O_\gamma(D_d^2 + bA_d) = d^{3+o_m(1)}.$$

In particular $c_d \leq d^{3+o_m(1)}$. Hence

$$n_d = \left\lfloor \frac{c_d}{\gamma b_d^2} \right\rfloor \leq d^M$$

for one fixed M and all sufficiently large dimensions in the subsequence. Moreover $c_d/(\gamma b_d^2) \geq 2$. The calibrated lower transfer now gives

$$\sup_{2 \leq T \leq d^M} \frac{\mathfrak{R}_\mathfrak{F}^B(d, T)}{\sqrt{T}} \geq \frac{\sqrt{\gamma}}{4} \sqrt{c_d}.$$

Therefore $\alpha_\mathfrak{F}^B \geq (\alpha_\mathfrak{F}^{\text{stop}} - 2\epsilon)/2$. Send $\epsilon \downarrow 0$ and use (18). $\square$

Remark 3.3 (From regret to a first hit). *For any n -round policy, let $p = \mathbb{P}_Q(\tau_b \leq n)$. The pre-hit inequality gives*

$$\mathbb{E}_Q R_n \geq \gamma bn(1 - p).$$

If $\mathbb{E}_Q R_n \leq \gamma bn/2$, then $p \geq 1/2$, and stopping the policy at $\tau_b \wedge n$ proves

$$\text{SC}_b(Q) \leq 2b \mathbb{E}_Q R_n.$$

Thus a uniform bound $D\sqrt{n} + A$ produces $\text{SC}_b(Q) = \tilde{O}(D^2 + bA)$ by choosing $n = \tilde{O}(D^2/b^2 + A/b)$. This is the reverse online-to-stop conversion needed in the exponent proof; it uses no terminal batch decision.

Corollary 3.4 (Full-class exponents). *For the full class of continuous $[0, 1]$ -valued convex losses,*

$$\frac{5}{2} \leq \alpha_{\text{full}}^{\text{stop}} \leq 3, \quad \frac{5}{4} \leq \alpha_{\text{full}}^{\text{reg}} \leq \frac{3}{2}. \quad (23)$$

Determining either exponent determines the other.

Proof. The upper endpoint is the current general BCO upper bound (Lattimore and György, 2023; Fokkema et al., 2024); the lower endpoint is the minimax theorem of Rajaraman (2026). Its 1-Lipschitz hard family embeds in the present bounded unit-noise model by the affine rescaling $g = (f + 1)/2$: from an observation $f(x) + \xi$, return $(f(x) + \xi + 1)/2 + \zeta$, where $\zeta \sim N(0, 3/4)$ is independent. The result is $g(x) + \xi'$ with $\xi' \sim N(0, 1)$, and regret changes only by a factor two. Apply Theorem 3.2. $\square$

4 Observation-adaptive continuation on the dimple family

Let $C = B_2^d(1)$. Fix

$$d^{-1} \leq \varepsilon \leq \varepsilon_0, \quad s = \sqrt{1 + 2\varepsilon},$$

where $\varepsilon_0 > 0$ is a sufficiently small constant. For $\theta \in S_2^{d-1}(1)$, define

$$F_\theta(x) = \varepsilon + \frac{1}{2}\|x\|^2 - \frac{1}{2}(s - \|\theta - x\|)_+^2. \quad (24)$$

These functions are continuous, convex, and $[0, 1]$ -valued after reducing ε_0 if necessary (Bakhtiari et al., 2025, Lemma 10). Their unique minimizers are θ , with $F_\theta(\theta) = 0$. At the origin,

$$b_0 := F_\theta(0) = s - 1 = \frac{2\varepsilon}{\sqrt{1 + 2\varepsilon} + 1} \asymp \varepsilon. \quad (25)$$

Use the rotation-invariant prior $\Theta \sim \text{Unif}(S_2^{d-1}(1))$. Since $\nabla F_\theta(0) = -b_0\theta$, convexity gives

$$F_\theta(x) \geq b_0\{1 - \langle \theta, x \rangle\}.$$

The origin is therefore the prior Bayes action and $r(Q_0) = b_0$. Write

$$\tau_\gamma = \inf\{t \geq 1 : r(Q_t) \leq \gamma b_0\}. \quad (26)$$

Definition 4.1 (Fresh-direction policies). *A policy is fresh-direction if, at round t , it first draws $B_t \sim \text{Unif}(S_2^{d-1}(1))$, independently of (Θ, H_{t-1}) , then chooses an arbitrary $\sigma(H_{t-1}, B_t)$ -measurable radius $R_t \in [0, 1]$ and queries $X_t = R_t B_t$.*

The radius may use the complete past. Only the direction must be fresh. The comparison does not cover all random-direction methods, such as paired probes centered at an adaptively chosen point.

Theorem 4.2 (Fresh-direction information barrier). *For $d \geq 2$, there is a universal $C < \infty$ such that, for every posterior ν of Θ , every fresh $B \sim \text{Unif}(S_2^{d-1}(1))$, every measurable $R = R(B) \in [0, 1]$, and independent $\xi \sim N(0, 1)$,*

$$I_\nu(\Theta; B, F_\Theta(RB) + \xi) \leq C(\varepsilon^4 + d^{-4}). \quad (27)$$

Under the spherical prior, every nonadaptive batch $X_1, \dots, X_n$, possibly randomized but independent of Θ and fixed before its observations, satisfies

$$I(\Theta; O_{1:n} \mid X_{1:n}) \leq Cn(\varepsilon^4 + d^{-4}). \quad (28)$$

Every fresh-direction stopping time N with $\mathbb{E}N < \infty$ satisfies

$$I(\Theta; H_N) \leq C(\varepsilon^4 + d^{-4}) \mathbb{E}N. \quad (29)$$

Conversely, if $\mathbf{M} \in \sigma(H_N)$ is an event on which the terminal posterior Bayes risk is at most γb_0 , then

$$I(\Theta; H_N) \geq \frac{d-1}{2}(1-\gamma)^2 \mathbb{P}(\mathbf{M}). \quad (30)$$

Thus a nonadaptive batch or a stopped fresh-direction experiment that contracts posterior risk with constant probability requires

$$\mathbb{E}N = \Omega_\gamma \left(\frac{d}{\varepsilon^4 + d^{-4}} \right). \quad (31)$$

Fix a sufficiently large $C_L = C_L(\gamma)$, and for $0 < a \leq b_0$ put

$$L_{d,a} = 1 + \log \left\{ \frac{C_L d (1 + \log(d/a))}{a} \right\}. \quad (32)$$

Theorem 4.3 (Observation-adaptive directional experiment). *There are constants $\gamma_0, c_0 > 0$, with $\gamma_0 + c_0 < \gamma$, and $C_\gamma < \infty$ such that, for every sufficiently large d , every $d^{-1} \leq \varepsilon \leq \varepsilon_0$, and every $0 < a \leq b_0$, an observation-adaptive experiment returns $\hat{x} \in B_2^d(1)$ with*

$$\sup_{\theta \in S_2^{d-1}(1)} \mathbb{P}_\theta \{F_\theta(\hat{x}) > \gamma_0 a\} \leq c_0 a, \quad (33)$$

$$N \leq \frac{C_\gamma d^2 L_{d,a}^2}{a^2}, \quad (34)$$

$$\sup_{\theta \in S_2^{d-1}(1)} \mathbb{E}_\theta R_N \leq \frac{C_\gamma d^2 L_{d,a}^2}{a}. \quad (35)$$

The query cap is deterministic. The conditional distributions and subspaces of the directions in the last two stages depend measurably on earlier observations. Under every prior on Θ ,

$$\mathbb{P} \{ \mathbb{E}[F_\Theta(\hat{x}) \mid H_N] \leq \gamma a \} \geq 1 - \frac{\gamma_0 + c_0}{\gamma} > 0. \quad (36)$$

For the next statement, $\text{SC}_b^{[1]}$ restricts (6) to one query, and $\text{SC}_b^{\text{fresh}}$ restricts it to fresh-direction policies.

Corollary 4.4 (Polynomial separations). *For all sufficiently large d , at $\varepsilon = d^{-1}$ and $b = b_0 \asymp d^{-1}$, the initial spherical state satisfies*

$$\text{SC}_b^{[1]} = \Omega_\gamma(d^3), \quad \text{SC}_b^{\text{fresh}} = \Omega_\gamma(d^3), \quad \text{SC}_b = \tilde{O}_\gamma(d^2). \quad (37)$$

At the same scale, a nonadaptive batch with constant contraction probability has size $\Omega_\gamma(d^5)$, and a stopped fresh-direction experiment has $\mathbb{E}N = \Omega_\gamma(d^5)$. The observation-adaptive construction uses $\tilde{O}_\gamma(d^4)$ queries.

Let

$$L_{d,T} = 1 + \log \{ C_L d \max\{T, d\} [1 + \log\{d \max\{T, d\}\}] \}.$$

Under the spherical prior, every fresh-direction policy satisfies

$$\mathbb{E}R_T \geq c_\gamma \min \left\{ \frac{T}{d}, d^4 \right\}, \quad (38)$$

while an unrestricted policy satisfies, uniformly in θ ,

$$\mathbb{E}_\theta R_T \leq C_\gamma L_{d,T} \min \left\{ \frac{T}{d}, d\sqrt{T} \right\}. \quad (39)$$

At $T = d^5$, these bounds are $\Omega(d^4)$ and $\tilde{O}(d^{7/2})$, respectively.

Proof of (37). For a one-query or fresh-direction first-hit experiment, let $p = \mathbb{P}(\mathbf{M})$. Theorem 4.2 gives

$$C(\varepsilon^4 + d^{-4}) \mathbb{E}N \geq I(\Theta; H_N) \geq c_\gamma d p.$$

Before the hit, Lemma 2.2 gives $\mathbb{E}R_N \geq \gamma b \mathbb{E}N$. Therefore

$$\frac{b \mathbb{E}R_N}{p} \geq \frac{c_\gamma b^2 d}{\varepsilon^4 + d^{-4}}.$$

At $b \asymp \varepsilon = d^{-1}$, this is $\Omega_\gamma(d^3)$. The same argument covers the one-query restriction.

Couple the capped routine from Theorem 4.3, run at $a = b_0$, to its first-hit truncation. The two policies agree until the hit, and (36) implies that the virtual capped path hits with probability at least $p_\gamma := 1 - (\gamma_0 + c_0)/\gamma > 0$. Truncation can only decrease regret, so

$$\frac{b \mathbb{E}R_{\tau_\gamma \wedge N}}{\mathbb{P}(\tau_\gamma \leq N)} \leq \frac{C_\gamma}{p_\gamma} b \frac{d^2 L_{d,b}^2}{b} = \tilde{O}_\gamma(d^2).$$

This proves the upper bound. $\square$

5 The fresh-direction barrier

Proof of Theorem 4.2. Put

$$q(x) = \varepsilon + \frac{1}{2} \|x\|^2.$$

For $B \in S_2^{d-1}(1)$, $r \in [0, 1]$, and $u = \langle \theta, B \rangle$,

$$|F_\theta(rB) - q(rB)| = \frac{1}{2} \left(s - \sqrt{1 + r^2 - 2ru} \right)_+^2.$$

Minimizing the square root over $r \in [0, 1]$ shows that the supremum of the right side is $(s - 1)^2/2$ when $u \leq 0$, and

$$\frac{1}{2} \left(s - \sqrt{1 - u^2} \right)^2$$

when $u > 0$. On $|u| \leq 1/2$, its square is at most $C(\varepsilon^4 + u^8)$; the complementary spherical cap has exponentially small probability. Rotation invariance and

$$\mathbb{E}_B \langle \theta, B \rangle^8 = \frac{105}{d(d+2)(d+4)(d+6)}$$

therefore give, uniformly in θ ,

$$\mathbb{E}_B \sup_{0 \leq r \leq 1} |F_\theta(rB) - q(rB)|^2 \leq C(\varepsilon^4 + d^{-4}). \quad (40)$$

Now condition on an arbitrary posterior ν . The fresh direction B is independent of Θ , and $R(B)$ is known after conditioning on B . Comparing the Gaussian observation channel with the Θ -independent reference $q(R(B)B) + \xi$, the information-radius bound gives

$$I_\nu(\Theta; B, F_\Theta(RB) + \xi) \leq \frac{1}{2} \mathbb{E}_{\Theta \sim \nu, B} |F_\Theta(RB) - q(RB)|^2.$$

Equation (40) proves (27).

For a fixed nonadaptive design point $x = ru$, rotation invariance interchanges a fixed direction u under uniform Θ with a uniform direction B under fixed θ . Thus (40) also gives

$$\mathbb{E}_\Theta |F_\Theta(x) - q(x)|^2 \leq C(\varepsilon^4 + d^{-4}).$$

Condition on a possibly randomized design and compare its joint Gaussian observation law with the product reference $(q(X_i) + \xi_i)_{i \leq n}$. Additivity of the reference KL divergence proves (28).

For a fresh-direction stopping time, let $N_n = N \wedge n$ and use the cemetery-padded transcript. Since $\{N \geq t\}$ is H_{t-1} -measurable, conditional application of (27) and the chain rule give

$$I(\Theta; H_{N_n}) \leq C(\varepsilon^4 + d^{-4}) \sum_{t=1}^n \mathbb{P}(N \geq t) = C(\varepsilon^4 + d^{-4}) \mathbb{E}N_n.$$

Continuity of relative entropy along the increasing stopped σ -fields, followed by $n \rightarrow \infty$, proves (29).

For the converse, convexity at the origin and $\nabla F_\theta(0) = -b_0\theta$ give

$$F_\theta(x) \geq b_0\{1 - \langle \theta, x \rangle\}. \quad (41)$$

With $m_N = \mathbb{E}[\Theta \mid H_N]$, taking the infimum over x in (41) gives directly

$$r(Q_{H_N}) \geq b_0(1 - \|m_N\|).$$

Hence $r(Q_{H_N}) \leq \gamma b_0$ on $\mathbf{M}$ implies $\|m_N\| \geq 1 - \gamma$. The uniform law σ on $S_2^{d-1}(1)$ is $1/(d-1)$ -sub-Gaussian. The variational formula for relative entropy therefore yields

$$D_{\text{KL}}(\nu \parallel \sigma) \geq \frac{d-1}{2} \|\mathbb{E}_\nu \Theta\|^2.$$

Apply this to each posterior and average on $\mathbf{M}$:

$$I(\Theta; H_N) = \mathbb{E}D_{\text{KL}}(Q_{H_N} \parallel \sigma) \geq \frac{d-1}{2} (1 - \gamma)^2 \mathbb{P}(\mathbf{M}).$$

Combining this inequality with (29) proves (31). $\square$

Proof of the lower bound in Corollary 4.4. Let $N = \tau_\gamma \wedge T$ and $p = \mathbb{P}(\tau_\gamma \leq T)$. Before the contraction time, every action has conditional expected regret at least $r(Q_{t-1}) > \gamma b_0$. Hence

$$\mathbb{E}R_T \geq \gamma b_0 \mathbb{E}N. \quad (42)$$

Moreover,

$$\mathbb{E}N \geq T(1 - p).$$

Apply Theorem 4.2 to the stopped transcript H_N . On $\{\tau_\gamma \leq T\}$, its terminal posterior risk is at most γb_0 , and therefore

$$Cd^{-4}\mathbb{E}N \geq I(\Theta; H_N) \geq c_\gamma dp$$

when $\varepsilon = d^{-1}$. Thus

$$\mathbb{E}N \geq c_\gamma d^5 p.$$

Minimizing $\max\{T(1 - p), c_\gamma d^5 p\}$ over $p \in [0, 1]$ gives $\mathbb{E}N \geq c_\gamma \min\{T, d^5\}$. Since $b_0 \asymp d^{-1}$, (42) proves (38). $\square$

6 The observation-adaptive experiment

The construction behind Theorem 4.3 has three stages. The table suppresses logarithmic factors.

stage	progress	(queries, regret)
orthogonal scan	obtain v_0 with $\langle \theta, v_0 \rangle = \tilde{\Omega}(\max\{\sqrt{\varepsilon}, d^{-1/2}\})$	$(d^2/\varepsilon^2, d^2/\varepsilon)$
tangent amplification	choose tangent directions from the current estimate and enter a fixed angular cone	$(d^2/\varepsilon^2, d^2/\varepsilon)$
angular contraction	invert boundary observations and repeat $q \mapsto q/2$ until $q = O(a)$	$(d^2/a^2, d^2/a)$

Only the last two stages orient later probes using earlier observations. The detailed proof is in Appendix A.

Proof of the upper statements in Corollary 4.4. The same adaptive experiment, run to accuracy a and followed by commitment, gives the family upper bound. At $a = b_0 \asymp \varepsilon$, Theorem 4.3 uses $\tilde{O}(d^2/\varepsilon^2)$ queries and $\tilde{O}(d^2/\varepsilon)$ regret. Equation (36) proves the adaptive part of the query separation. Since $\varepsilon^4 + d^{-4} \leq 2\varepsilon^4$ in the stated range, Theorem 4.2 proves the lower part for both nonadaptive and fresh-direction experiments. Specializing to $\varepsilon = d^{-1}$ gives the displayed endpoint.

For the regret upper bound, first note that playing the origin for all T rounds costs $b_0 T \leq CT/d$. When $T \geq C_\gamma d^4 L_{d,T}^2$, choose

$$a = A_\gamma \frac{dL_{d,T}}{\sqrt{T}}$$

with A_γ large enough. Then $a \leq b_0$, $L_{d,a} \leq CL_{d,T}$, and (34) uses at most $T/2$ rounds. Indeed, enlarging the crossover constant makes $a \leq c/d \leq b_0$, while $1/a \leq \sqrt{T}/d$ and $\log(d/a) \leq \log(dT)$ give the displayed logarithmic comparison directly from (32). The exploration regret is at most

$$\frac{C_\gamma d^2 L_{d,a}^2}{a} \leq C_\gamma d L_{d,T} \sqrt{T}.$$

Play $\hat{x}$ for the remaining rounds. Since losses are bounded by one, (33) gives

$$\sup_{\theta} \mathbb{E}_{\theta} F_{\theta}(\hat{x}) \leq (\gamma_0 + c_0)a \leq C_\gamma a.$$

The exploitation regret is therefore at most $C_\gamma a T \leq C_\gamma d L_{d,T} \sqrt{T}$. Below the displayed crossover, use the origin strategy. If $T \leq d^4$, its cost CT/d is the smaller term in (39). If $d^4 < T < C_\gamma d^4 L_{d,T}^2$, then

$$\frac{T}{d} \leq C_\gamma L_{d,T} d \sqrt{T}.$$

These two cases prove (39) throughout the remaining range. $\square$

7 Scope and open problem

The separation proves that adaptive radii do not overcome the barrier when each direction remains fresh and Haar. The successful construction instead uses paired, structured probes whose later directions depend on earlier observations. In the Bellman recursion (14), this distinction is carried by the posterior continuation: a weak first observation is useful because it changes the directions available to the rest of the experiment.

The full-class BCO gap remains open. Our lower bounds cover nonadaptive batches and fresh radial directions, not general adaptive directional policies, and the unrestricted dimple-family upper bound has no matching lower bound. By Theorem 3.2 and the full-class lower bound of [Rajaraman \(2026\)](#), closing the regret gap is equivalent to deciding whether the stopped-complexity exponent lies at the lower endpoint $5/2$, the upper endpoint 3 , or strictly between them.

Open question 7.1 (Full-class stopped complexity). *Determine $\alpha_{\text{full}}^{\text{stop}}$ for the full bounded convex class. Equivalently, determine the dimension exponent of stochastic BCO regret. A constructive upper bound requires a uniformly low-cost posterior experiment; a lower bound may use Le Cam, Fano, or geometric testing, but must rule out the full observation-adaptive continuation class.*

A Proof of the observation-adaptive experiment

Throughout this appendix, fix $0 < a \leq b_0$ and total failure probability $\delta = c_\delta a$, where $c_\delta > 0$ is fixed below, and put

$$\mathcal{L} = 1 + \log\{16d(1 + \log(d/a))/\delta\}.$$

This agrees with $L_{d,a}$ after increasing C_γ . Fix a sufficiently large universal C_{geo} , then choose $q_\star > 0$ so that

$$q_\star < s/4, \quad C_{\text{geo}}q_\star \leq s/2$$

uniformly over the stated range of ε . Next choose $\gamma_0, c_\delta > 0$ so that $\gamma_0 + c_\delta < \gamma$, and set $c_0 = c_\delta$, followed by the small geometric steps, termination tolerances, and large repetition constants used below. All constants are fixed in this order.

We repeatedly use the standard consequence of sphere concentration that, for a Haar orthonormal basis (u_i) in a fixed k -dimensional subspace and a fixed unit vector w ,

$$\max_i |\langle w, u_i \rangle| \leq C\sqrt{\mathcal{L}/k} \quad (43)$$

outside probability $e^{-c\mathcal{L}}$ when $\mathcal{L} \leq k$; when $\mathcal{L} > k$, we use the deterministic bound one. Every repetition below uses fresh Gaussian noise. Each stage has a deterministic query cap, the number of stages is at most $C(1 + \log(d/a))$, and confidence levels are divided across these caps so that the union of all concentration failures is at most δ .

Every use of Haar concentration is conditional on the transcript before the corresponding basis is drawn. At that point, the current iterate, the hidden direction, and the activity margin are fixed, while the new basis is Haar in its tangent space. Iterated conditioning and the allocated union bound give a good event $\mathcal{H}$ on which all activity, estimation, and contraction invariants below hold. The deterministic caps define the routine on $\mathcal{H}^c$. The procedure is made total and Borel by returning the origin whenever a normalized estimate is undefined and by resolving every gate equality with the stopping branch.

Whenever $r = \|\theta - x\| < s$, expanding (24) gives

$$F_\theta(x) = s\|\theta - x\| + \langle \theta, x \rangle - 1. \quad (44)$$

A.1 Orthogonal initialization

Lemma A.1. *With probability at least $1 - \delta/3$, the construction below returns a unit vector v_0 satisfying*

$$\langle \theta, v_0 \rangle \geq c \min \left\{ 1, \max \left(\sqrt{\varepsilon}, \sqrt{\mathcal{L}/d} \right) \right\}. \quad (45)$$

It uses at most $Cd^2\mathcal{L}^2/\varepsilon^2$ queries and incurs at most $Cd^2\mathcal{L}^2/\varepsilon$ regret on the success event.

Proof. Draw a Haar basis $(u_i)_{i=1}^d$, and set

$$h_0 = c_h \min\{\sqrt{\varepsilon}, \varepsilon\sqrt{d/\mathcal{L}}\}, \quad n_0 = \lceil Cd\mathcal{L}^2/\varepsilon^2 \rceil. \quad (46)$$

Estimate every centered difference

$$D_i^0 = \frac{F_\theta(h_0 u_i) - F_\theta(-h_0 u_i)}{2h_0}$$

using n_0 repetitions at each of the two points. On (43),

$$h_0^2 + 2h_0 \max_i |\langle \theta, u_i \rangle| < 2\varepsilon$$

for sufficiently small c_h , so both points lie in the active dimple. Equation (44) gives

$$D_i^0 = -\lambda_i \langle \theta, u_i \rangle, \quad \lambda_i = \frac{2s}{r_i^+ + r_i^-} - 1, \quad |\lambda_i - b_0| \leq Ch_0^2, \quad (47)$$

where $r_i^\pm = \|\theta \mp h_0 u_i\|$. Hence the noiseless difference vector is

$$D = -b_0 \theta + e_0, \quad \|e_0\| \leq Ch_0^2.$$

The Gaussian estimation error ζ is isotropic with coordinate variance $(2n_0 h_0^2)^{-1}$. At the allocated joint failure level,

$$|\langle \theta, \zeta \rangle| \leq \frac{C\varepsilon}{h_0 \sqrt{d\mathcal{L}}}, \quad (48)$$

$$\|\zeta\| \leq \frac{C\varepsilon}{h_0 \mathcal{L}} \left(1 + \sqrt{\mathcal{L}/d}\right). \quad (49)$$

Let

$$t = \min\{1, \max(\sqrt{\varepsilon}, \sqrt{\mathcal{L}/d})\}.$$

When $\mathcal{L} \leq d$, the definition of h_0 gives $h_0 \asymp c_h \varepsilon / t$. After choosing c_h small and then the repetition constant large,

$$\langle \theta, -\widehat{D} \rangle \geq c\varepsilon, \quad \|\widehat{D}\| \leq C\varepsilon/t.$$

When $\mathcal{L} > d$, one has $t = 1$; the same conclusions follow from (49), $\varepsilon \geq d^{-1}$, and the definition of h_0 . Set

$$v_0 = -\widehat{D}/\|\widehat{D}\|.$$

Then (45) holds.

The stage uses $2dn_0 = O(d^2 \mathcal{L}^2 / \varepsilon^2)$ queries. Every query point has loss $O(\varepsilon)$, so its regret is $O(d^2 \mathcal{L}^2 / \varepsilon)$. $\square$

A.2 Tangent amplification

Lemma A.2. *Conditionally on the initialization event, a capped tangent routine reaches $\|\theta - v\| \leq q_\star$ with failure probability at most $\delta/3$. If c_{init} denotes the right side of (45), the stage uses*

$$N_{\text{amp}} \leq \frac{Cd^2 \mathcal{L}}{c_{\text{init}}^4}, \quad R_{\text{amp}} \leq \frac{Cd^2 \mathcal{L}}{c_{\text{init}}^2}, \quad (50)$$

up to lower-order $C\mathcal{L}(1 + \log(1/c_{\text{init}}))$ gate costs.

Proof. Suppose a unit vector v satisfies $\langle \theta, v \rangle \geq c$, where

$$c_{\text{init}} \leq c \leq 1 - q_\star^2/8.$$

The constants ensure $c^2 \geq c_1 \varepsilon$. On the unit sphere, $F_\theta(v)$ is the nondecreasing function

$$\phi_\varepsilon(q) = \varepsilon + \frac{1}{2} - \frac{1}{2}(s - q)_+^2, \quad q = \|\theta - v\|.$$

The fixed choice of $q_\star$ makes

$$\phi_\varepsilon(q_\star) - \phi_\varepsilon(q_\star/2)$$

uniformly positive. Averaging $O(\mathcal{L})$ repeated observations at v therefore separates $q \leq q_*/2$ from $q \geq q_*$, with either decision allowed in the overlap. Stopping implies $q \leq q_*$, while continuation implies $q \geq q_*/2$.

On a continuing stage, put $x = (c/2)v$ and draw a Haar tangent basis $(u_i)_{i=1}^{d-1}$ of $v^\perp$. Write

$$p = \langle \theta, v \rangle, \quad r_0 = \|\theta - x\|.$$

Then

$$r_0^2 = 1 + \frac{c^2}{4} - cp \leq 1 - \frac{3c^2}{4}, \quad r_0 \geq 1 - \frac{c}{2}. \quad (51)$$

When $\mathcal{L} \leq d$, take $h = c_2 c$. On $\mathcal{H}$, (43) and $c \geq c_{\text{init}} \gtrsim \sqrt{\mathcal{L}/d}$ give

$$|\langle \theta, u_i \rangle| \leq Cc.$$

Hence $x \pm hu_i$ remain in the active dimple for small enough c_2 . When $\mathcal{L} > d$, take $h = c_2 c^2$ and use the deterministic bound one; here c is bounded below by a universal positive constant, so the same conclusions hold after changing constants.

Let $q_i = \langle \theta, u_i \rangle$. The exact centered differences are

$$D_i = -\lambda_i q_i, \quad \lambda_i = \frac{2s}{r_i^+ + r_i^-} - 1. \quad (52)$$

Writing $\lambda_0 = s/r_0 - 1$, a second-order Taylor expansion of $\sqrt{r_0^2 + t}$ at $t_\pm = h^2 \mp 2hq_i$ gives

$$c_3 c^2 \leq \lambda_0 \leq Cc, \quad |\lambda_i - \lambda_0| \leq Ch^2.$$

Put

$$g = -\sum_i D_i u_i = \sum_i \lambda_i q_i u_i.$$

Because the continuation gate ensures $\|\theta - v\| \geq q_*/2$, the tangent component of θ is bounded below. Choosing c_2 sufficiently small gives

$$\langle \theta, g \rangle \geq c_4 c^2, \quad \|g\| \leq Cc. \quad (53)$$

Estimate every D_i with

$$n_c = \lceil Cd\mathcal{L}/c^4 \rceil$$

repetitions at each side. Gaussian concentration in $v^\perp$ gives an error vector ζ satisfying

$$\|\zeta\| \leq c_{\text{amp}} c, \quad |\langle \theta, \zeta \rangle| \leq (c_4/2)c^2$$

on the joint event. More explicitly, with empirical differences $\widehat{D}_i$, set

$$\widehat{g} := -\sum_i \widehat{D}_i u_i = g + \zeta, \quad u := \widehat{g}/\|\widehat{g}\|.$$

Then $u \in v^\perp$ and

$$\langle \theta, u \rangle \geq k_0 c$$

for a universal $k_0 > 0$. Choose a fixed $0 < k \leq \min\{k_0, q_*/4\}$, and replace

$$v \quad \text{by} \quad \frac{v + ku}{\sqrt{1 + k^2}}.$$

Its correlation lower bound increases from c to at least $\sqrt{1+k^2}c$. There are $O(\log(2/c_{\text{init}}))$ stages.

At correlation c , the stage uses $O(d^2\mathcal{L}/c^4)$ queries. Each query has loss $O(\varepsilon + c^2) = O(c^2)$. Geometric summation gives (50). Since

$$c_{\text{init}} \gtrsim \max\{\sqrt{\varepsilon}, \sqrt{\mathcal{L}/d}\}$$

unless it is already a fixed constant, these costs are at most

$$O(d^2\mathcal{L}/\varepsilon^2) \quad \text{and} \quad O(d^2\mathcal{L}/\varepsilon),$$

respectively. □

A.3 Angular contraction

Lemma A.3. *Conditionally on the first two stage events, the capped contraction stage returns $\hat{x}$ with $F_\theta(\hat{x}) \leq \gamma_0 a$ outside probability $\delta/3$. It has deterministic query cap*

$$Cd^2\mathcal{L}/a^2$$

and success-event regret at most

$$Cd^2\mathcal{L}/a.$$

Proof. Maintain the invariant

$$\|\theta - v_j\| \leq Q_j, \quad Q_j = q_* 2^{-j}.$$

At one step, write $v = v_j$, $Q = Q_j$, take $h = c_5 Q$, and draw a Haar tangent basis (u_i) of $v^\perp$. Define

$$\alpha_h = (1 + h^2)^{-1/2}, \quad t = h(1 + h^2)^{-1/2}, \quad x_i^\pm = \alpha_h v \pm t u_i.$$

These points lie on the unit sphere and within distance CQ of θ , hence in the active dimple.

Write

$$\theta = c_v v + \sum_i q_i u_i, \quad c_v = 1 - \frac{1}{2} \|\theta - v\|^2,$$

and let

$$\psi(r) = sr - \frac{1}{2}r^2, \quad G(p) = \psi(\sqrt{2-2p}).$$

Equation (44) gives the exact inversion

$$q_i = \frac{G^{-1}(F_\theta(x_i^+)) - G^{-1}(F_\theta(x_i^-))}{2t}. \tag{54}$$

All inner products in this display belong to

$$I_Q = \{p \in [-1, 1] : \sqrt{2-2p} \leq C_{\text{geo}} Q\}.$$

On this interval, G is strictly monotone and

$$|(G^{-1})'(G(p))| = \frac{\sqrt{2-2p}}{s - \sqrt{2-2p}} \leq CQ.$$

The inverse derivative extends continuously to $p = 1$.

Project each empirical function value onto $G(I_Q)$ before applying G^{-1} . Estimate each value to accuracy $c_6 Q/\sqrt{d}$, using

$$\lceil Cd\mathcal{L}/Q^2 \rceil$$

repetitions per site. Every recovered tangent coefficient in (54) then has error at most $Cc_6 Q/\sqrt{d}$. If

$$w = \sum_i q_i u_i, \quad \hat{w} = \sum_i \hat{q}_i u_i,$$

then $\|\hat{w} - w\| \leq Cc_6 Q$. Clip $\hat{w}$ to norm at most $1/2$, and set

$$v_{\text{new}} = \sqrt{1 - \|\hat{w}\|^2} v + \hat{w}.$$

The map $w \mapsto \sqrt{1 - \|w\|^2} v + w$ is 2-Lipschitz on this ball. Choosing c_6 sufficiently small yields

$$\|\theta - v_{\text{new}}\| \leq Q/2.$$

At scale Q , the query count is

$$O(d^2 \mathcal{L}/Q^2).$$

Every query has loss $\psi(O(Q)) = O(Q)$, so the regret is

$$O(d^2 \mathcal{L}/Q).$$

Geometric summation until $Q \leq c_{\text{term}} a$ proves the displayed query and regret bounds. For sufficiently small c_{term} ,

$$F_\theta(\hat{x}) = \psi(\|\theta - \hat{x}\|) \leq \gamma_0 a.$$

□

Proof of Theorem 4.3. Run Lemmas A.1–A.3 in order, assigning failure probability $\delta/3$ to each. Because $a \leq b_0 \asymp \varepsilon$, their deterministic caps and geometric sums give

$$N \leq C_\gamma d^2 L_{d,a}^2 / a^2$$

and, on $\mathcal{H}$,

$$R_N \leq C_\gamma d^2 L_{d,a}^2 / a.$$

On $\mathcal{H}^c$, the capped routine uses no more than its displayed query budget and incurs at most one unit of regret per query. Since $\mathbb{P}(\mathcal{H}^c) \leq \delta = c_\delta a$, this adds at most

$$C_\gamma d^2 L_{d,a}^2 / a$$

to expected regret. The accuracy statement follows from Lemma A.3 and the union bound. Since losses are bounded by one, the same statement gives, under every prior,

$$\mathbb{E}F_\Theta(\hat{x}) \leq (\gamma_0 + c_0)a.$$

Apply Markov's inequality to the nonnegative random variable $\mathbb{E}[F_\Theta(\hat{x}) \mid H_N]$ to obtain (36). On the displayed event, the posterior Bayes risk is no larger than the posterior loss of $\hat{x}$, and hence is at most γa . □

B Measurable implementation and Bayes–minimax duality

This appendix proves Theorem 3.1. It establishes measurable implementation and the order of the Bayes and minimax optimizations; the two-sided stopped reduction itself was proved in Section 2.

Lemma B.1 (Prior-wise Borel realization). *Fix a Borel prior Q and a finite horizon T . On standard Borel history, action, and observation spaces, every universally measurable randomized nonanticipating policy has a Borel randomized nonanticipating policy inducing the same joint law of (F, H_T) under Q . The same holds when the action space is augmented by a stopping symbol.*

Proof. At time t , represent the action kernel as a universally measurable map from histories into the standard Borel space of action distributions. Under the reached history law, this map has a Borel version. Replace it by that version; the strategic law through time t is unchanged. Induction over $t = 1, \dots, T$ proves the claim. Adding one isolated stopping symbol preserves the standard Borel property. $\square$

B.1 Finite-horizon Bayes–minimax identity

Proof of (18). Let $\nu = N(0, 1)$, let $\mathbf{H}_T = (C \times \mathbb{R})^T$, and let $\mathcal{P}_T$ be the set of reference strategic measures

$$P(dh_T) = \prod_{t=1}^T \pi_t(dx_t \mid h_{t-1}) \nu(dy_t)$$

generated by Borel policies. The set $\mathcal{P}_T$ is convex and weakly compact. Indeed, C is compact and every member has $\nu^{\otimes T}$ as its observation marginal, so the family is tight. It is closed because the conditional-independence identities

$$\int a(h_{t-1}, x_t) b(y_t) dP = \left(\int b d\nu \right) \int a(h_{t-1}, x_t) dP$$

for bounded continuous a, b pass to weak limits. Standard-Borel disintegration identifies each limiting strategic measure with a Borel nonanticipating policy.

For $f \in \mathfrak{F}_d(C)$, put

$$L_f(h_T) = \exp \left\{ \sum_{t=1}^T \left(y_t f(x_t) - \frac{1}{2} f(x_t)^2 \right) \right\}$$

and

$$u(P, f) = \int L_f(h_T) \sum_{t=1}^T \{f(x_t) - \min_C f\} dP(h_T).$$

Gaussian change of measure gives $u(P, f) = \mathbb{E}_f^{\pi} R_T$. The payoff is affine and continuous in P , and continuous in f under the uniform topology. Continuity follows by domination with

$$T \exp \left\{ \sum_{t=1}^T |y_t| \right\},$$

which is integrable uniformly over $\mathcal{P}_T$.

Let $\mathcal{Q}_0$ be the convex set of finitely supported priors on $\mathfrak{F}_d(C)$, and set $u(P, Q) = \int u(P, f) Q(df)$. Sion's theorem (Sion, 1958), with compact side $\mathcal{P}_T$, yields

$$\min_{P \in \mathcal{P}_T} \sup_f u(P, f) = \sup_{Q \in \mathcal{Q}_0} \min_{P \in \mathcal{P}_T} u(P, Q).$$

Allowing all Borel priors cannot decrease the right-hand side, and weak duality prevents it from exceeding the left-hand side. Taking the outer supremum over C proves (18). $\square$

B.2 Measurable selection of stopped experiments

Lemma B.2 (Capped Gaussian selection). *Fix a query cap $L < \infty$. The capped stopped value*

$$v_L(Q, b) = \inf_{\varepsilon: N \leq L} \frac{b \mathbb{E}_Q R_N}{\mathbb{P}_Q(\mathbf{M})}$$

is universally measurable. For every $\epsilon > 0$, there is a universally measurable state-to-experiment rule with value at most $v_L(Q, b) + \epsilon$ at every active state.

Proof. The belief space $\mathcal{P}(\mathfrak{F}_d(C))$ is standard Borel. For a belief μ , define

$$\ell(\mu, x) = \int \{f(x) - \min_C f\} \mu(df), \quad r(\mu) = \min_{x \in C} \ell(\mu, x).$$

The first map is Borel and continuous in x ; compactness of C makes r Borel and supplies a Borel Bayes-action selector. With φ the standard Gaussian density, the posterior update is the Borel map

$$\Psi(\mu, x, o)(df) = \frac{\varphi(o - f(x)) \mu(df)}{\int \varphi(o - f'(x)) \mu(df')}.$$

For a rational price $q > 0$, the capped penalized problem is exactly the recursion (14), with $\mu' = \Psi(\mu, x, o)$. Its one-step costs are bounded below and all primitive kernels are Borel. Standard integration, partial minimization, and universally measurable selection for lower-semianalytic dynamic programs therefore apply (Bertsekas and Shreve, 1978, Propositions 7.47–7.50). In particular, its root value $J_L^q(\mu, b)$ is lower semianalytic and has universally measurable near-minimizers.

Directly from the definition,

$$v_L(\mu, b) < q \iff J_L^q(\mu, b) < 0.$$

The rational strict sublevel sets make v_L universally measurable. On $\{v_L < +\infty\}$, enumerate the positive rationals and measurably choose the first q with

$$v_L + \epsilon/4 < q < v_L + \epsilon/2.$$

Then $J_L^q < 0$. Partition this set further by the first integer n such that $J_L^q \leq -1/n$, and use a Bellman selector whose approximation error is less than $1/(2n)$. The resulting penalized value remains negative, so its success probability is positive and its ratio is below $q < v_L + \epsilon/2$. On $\{v_L = +\infty\}$, any measurable one-query rule satisfies the extended-value approximation. Countable patching gives one universally measurable ϵ -optimal rule. $\square$

Lemma B.3 (Removing the query cap). *The unrestricted value $\text{SC}_b(Q)$ is universally measurable and has a universally measurable ϵ -optimal selector for every $\epsilon > 0$. The selected experiment may have a finite deterministic cap $L(Q, b)$ depending on the state.*

Proof. Truncate an admissible experiment after L queries. Its stopped regret and hit probability increase to those of the complete experiment:

$$\mathbb{E} R_{N \wedge L} \uparrow \mathbb{E} R_N, \quad \mathbb{P}(\mathbf{M}_L) \uparrow \mathbb{P}(\mathbf{M}).$$

Hence its ratio converges, including the zero-success case under the $+\infty$ convention, and

$$\text{SC}_b(Q) = \inf_{L \geq 1} v_L(Q, b).$$

This is universally measurable by Lemma B.2. On $\{\text{SC}_b < +\infty\}$, choose measurably the first L for which $v_L < \text{SC}_b + \epsilon/2$, then use the capped $\epsilon/2$ -selector. On $\{\text{SC}_b = +\infty\}$, use any measurable one-query rule; the approximation statement is vacuous there. Countable patching proves the claim. $\square$

Proof of (19). Lemma B.3 selects, simultaneously at every active state, an experiment within ϵ of its pointwise infimum. Taking the worst state and then letting $\epsilon \downarrow 0$ gives $\mathbf{SC}^{\text{unif}} \leq \mathbf{SC}^{\text{pt}}$. The reverse inequality is immediate from the definitions. For each fixed prior and finite global horizon, Lemma B.1 replaces the universal selector by a Borel policy with the same strategic law. $\square$

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References

- Dilip Arumugam and Benjamin Van Roy. Deciding what to learn: A rate-distortion approach. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pages 373–382. PMLR, 2021. URL <https://proceedings.mlr.press/v139/arumugam21a.html>.
- Alireza Bakhtiari, Tor Lattimore, and Csaba Szepesvári. Thompson sampling for bandit convex optimisation. In *Proceedings of the 38th Conference on Learning Theory*, volume 291 of *Proceedings of Machine Learning Research*, pages 231–263. PMLR, 2025. URL <https://proceedings.mlr.press/v291/bakhtiari25a.html>.
- Dimitri P. Bertsekas and Steven E. Shreve. *Stochastic Optimal Control: The Discrete-Time Case*. Academic Press, New York, 1978. URL https://web.mit.edu/dimitrib/www/SOC_1978.pdf.
- Dimitri P. Bertsekas and John N. Tsitsiklis. An analysis of stochastic shortest path problems. *Mathematics of Operations Research*, 16(3):580–595, 1991. doi: 10.1287/moor.16.3.580. URL <https://pubsonline.informs.org/doi/abs/10.1287/moor.16.3.580>.
- Sébastien Bubeck and Mark Sellke. First-order Bayesian regret analysis of Thompson sampling. *IEEE Transactions on Information Theory*, 69(3):1795–1823, 2023. URL <https://ieeexplore.ieee.org/document/9915622/>.
- Alexandra Carpentier. A simple and improved algorithm for noisy, convex, zeroth-order optimisation. *Mathematical Statistics and Learning*, 8:165–192, 2025. doi: 10.4171/MSL/54. URL <https://ems.press/content/serial-article-files/51581>.
- Fan Chen, Dylan J. Foster, Yanjun Han, Jian Qian, Alexander Rakhlin, and Yunbei Xu. Assouad, Fano, and Le Cam with interaction: A unifying lower bound framework and characterization for bandit learnability. In *Advances in Neural Information Processing Systems*, volume 37, 2024. doi: 10.52202/079017-2407. URL https://proceedings.neurips.cc/paper_files/paper/2024/hash/8a23a95e26d016711c0d70f79ade3c95-Abstract-Conference.html.
- Herman Chernoff. Sequential design of experiments. *The Annals of Mathematical Statistics*, 30:755–770, 1959. doi: 10.1214/aoms/1177706205. URL <https://doi.org/10.1214/aoms/1177706205>.
- Shi Dong and Benjamin Van Roy. An information-theoretic analysis for Thompson sampling with many actions. In *Advances in Neural Information Processing Systems 31*, pages 4161–4169, 2018. URL <https://arxiv.org/abs/1805.11845>.
- J. H. Eaton and Lotfi A. Zadeh. Optimal pursuit strategies in discrete-state probabilistic systems. *Journal of Basic Engineering*, 84(1):23–29, 1962. URL <https://asmedigitalcollection.asme.org/fluidsengineering/article-abstract/84/1/23/424641/Optimal-Pursuit-Strategies-in-Discrete-State?redirectedFrom=PDF>.
- Hidde Fokkema, Dirk van der Hoeven, Tor Lattimore, and Jack J. Mayo. Online Newton method for bandit convex optimisation. In *Proceedings of Thirty Seventh Conference on Learning Theory*, volume 247 of *Proceedings of Machine Learning Research*, pages 1713–1714. PMLR, 2024. URL <https://proceedings.mlr.press/v247/fokkema24a.html>. Full version: arXiv:2406.06506.
- Dylan J. Foster, Sham M. Kakade, Jian Qian, and Alexander Rakhlin. The statistical complexity of interactive decision making, 2021. URL <https://arxiv.org/abs/2112.13487>.

- Dylan J. Foster, Alexander Rakhlin, Ayush Sekhari, and Karthik Sridharan. On the complexity of adversarial decision making. In *Advances in Neural Information Processing Systems*, volume 35, pages 35404–35417, 2022. URL https://proceedings.neurips.cc/paper_files/paper/2022/hash/e5daf532498ebba4fe6544d49c811678-Abstract-Conference.html.
- Dylan J. Foster, Noah Golowich, and Yanjun Han. Tight guarantees for interactive decision making with the decision-estimation coefficient. In *Proceedings of the 36th Conference on Learning Theory*, volume 195 of *Proceedings of Machine Learning Research*, pages 3969–4043. PMLR, 2023. URL <https://proceedings.mlr.press/v195/foster23b.html>.
- Amaury Gouverneur, Borja Rodríguez-Gálvez, Tobias J. Oechtering, and Mikael Skoglund. Chained information-theoretic bounds and tight regret rate for linear bandit problems, 2024. URL <https://arxiv.org/abs/2403.03361>.
- Tor Lattimore and András György. Mirror descent and the information ratio. In *Proceedings of the 34th Conference on Learning Theory*, volume 134 of *Proceedings of Machine Learning Research*, pages 2965–2992. PMLR, 2021. URL <https://proceedings.mlr.press/v134/lattimore21b.html>.
- Tor Lattimore and András György. A second-order method for stochastic bandit convex optimisation. In *Proceedings of the 36th Conference on Learning Theory*, volume 195 of *Proceedings of Machine Learning Research*, pages 2067–2094. PMLR, 2023. URL <https://proceedings.mlr.press/v195/lattimore23a.html>.
- Shaojie Li and Yunbei Xu. Pointwise generalization in deep neural networks, 2026. URL <https://arxiv.org/abs/2605.18598>.
- Arnab Maiti, Yunbei Xu, and Kevin Jamieson. On the power of adaptivity for ε -best arm identification in linear bandits. In *Proceedings of the Thirty-Ninth Conference on Learning Theory*, volume 336 of *Proceedings of Machine Learning Research*, pages 4911–4968. PMLR, 2026. URL <https://proceedings.mlr.press/v336/maiti26a.html>.
- Mohammad Naghshvar and Tara Javidi. Active sequential hypothesis testing. *The Annals of Statistics*, 41(6):2703–2738, 2013. doi: 10.1214/13-AOS1144. URL <https://doi.org/10.1214/13-AOS1144>.
- Gergely Neu, Matteo Papini, and Ludovic Schwartz. Optimistic information directed sampling. In *Proceedings of the 37th Conference on Learning Theory*, volume 247 of *Proceedings of Machine Learning Research*, pages 3970–4006. PMLR, 2024. URL <https://proceedings.mlr.press/v247/neu24a.html>.
- Nived Rajaraman. The price of hidden curvature: An $\tilde{\Omega}(d^{5/4}\sqrt{T})$ lower bound for bandit convex optimization, 2026. URL <https://arxiv.org/abs/2607.18652>.
- Daniel Russo and Benjamin Van Roy. An information-theoretic analysis of Thompson sampling. *Journal of Machine Learning Research*, 17(68):1–30, 2016. URL <https://jmlr.org/papers/v17/14-087.html>.
- Daniel Russo and Benjamin Van Roy. Learning to optimize via information-directed sampling. *Operations Research*, 66(1):230–252, 2018. doi: 10.1287/opre.2017.1663. URL <https://pubsonline.informs.org/doi/10.1287/opre.2017.1663>.

- Daniel Russo and Benjamin Van Roy. Satisficing in time-sensitive bandit learning. *Mathematics of Operations Research*, 47(4):2815–2839, 2022. doi: 10.1287/moor.2021.1229. URL <https://doi.org/10.1287/moor.2021.1229>.
- Daniel J. Russo, Benjamin Van Roy, Abbas Kazerouni, Ian Osband, and Zheng Wen. A tutorial on thompson sampling. *Foundations and Trends in Machine Learning*, 11(1):1–96, 2018. doi: 10.1561/22000000070. URL <https://www.nowpublishers.com/article/Details/MAL-070>.
- Maurice Sion. On general minimax theorems. *Pacific Journal of Mathematics*, 8(1):171–176, 1958. doi: 10.2140/pjm.1958.8.171. URL <https://doi.org/10.2140/pjm.1958.8.171>.
- Yunbei Xu. Bellman-sufficient information complexity, 2026a. URL <https://arxiv.org/abs/2606.11171>.
- Yunbei Xu. The dimension of nonterminating resampling computations. *arXiv preprint arXiv:2607.17469*, 2026b. doi: 10.48550/arXiv.2607.17469. URL <https://arxiv.org/abs/2607.17469>.
- Yunbei Xu. Pointwise complexity for Gaussian fields: Upper envelopes, algorithmic lower bounds, and separation, 2026c. URL <https://arxiv.org/abs/2606.07931>.
- Yunbei Xu and Assaf Zeevi. Bayesian design principles for frequentist sequential learning. *Journal of the ACM*, 72(5):37:1–37:65, October 2025. doi: 10.1145/3766898. URL <https://dl.acm.org/doi/10.1145/3766898>.